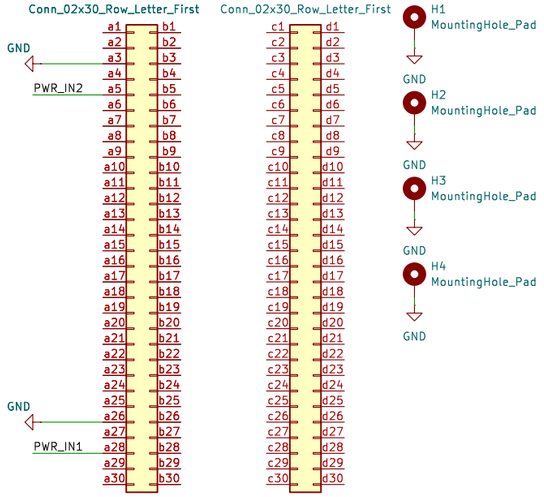
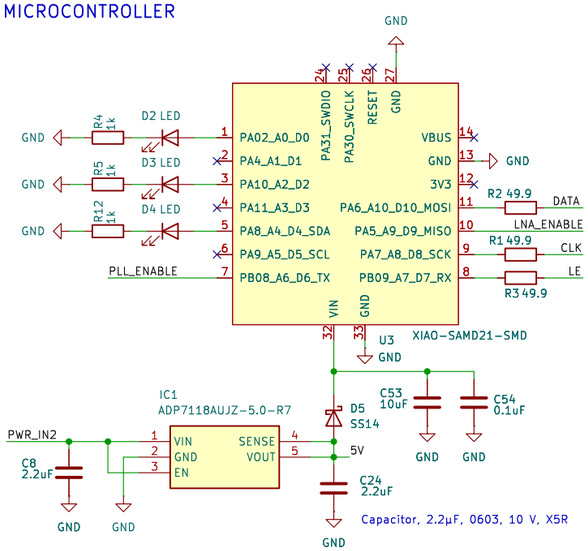
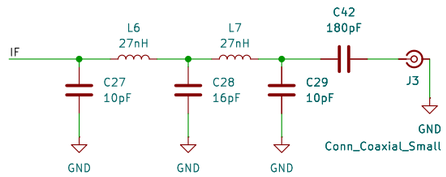
PIN OUT



MICROCONTROLLER



FILTER

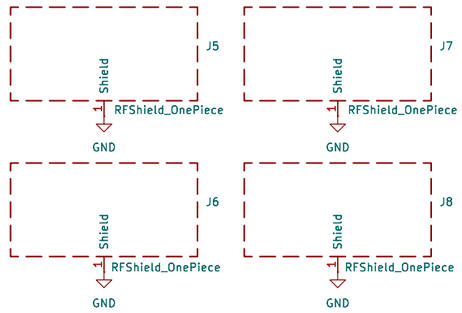


Low-pass filter, 440MHz cutoff
5th order
Chebyshev , PI-type,
Impedance 50

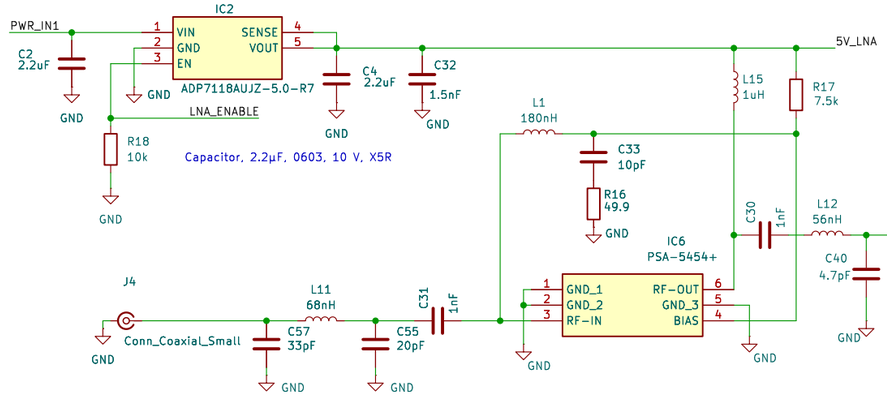
Filter Components:
Dielectric NPO/COG
Case 0402
Tolerance Cap 1% / Ind 2%
SRF 3.5GHz
GJM1555C1H100FB01
GJM1555C1H160FB01
LQW15AN27NG00

PCB Stackup & Fabrication ParametersLayer Count:
4Board Thickness: 1.6 mm
Base Material: FR4Layer Order: Top (RF/Signal) | In1 (GND) | In2 (VCC) | Bottom (Signal)
Outer Copper Weight: 1 oz (0.035 mm)
Inner Copper Weight: 0.5 oz (0.015 mm)
RF Dielectric (Top to In1): 0.2 mm (Prepreg 7628, Er: approx 4.4)RF
JLCPCB Profile: JLC04161H-7628 (Standard)
Impedance Control Service: No
NoSurface Finish: ENIG

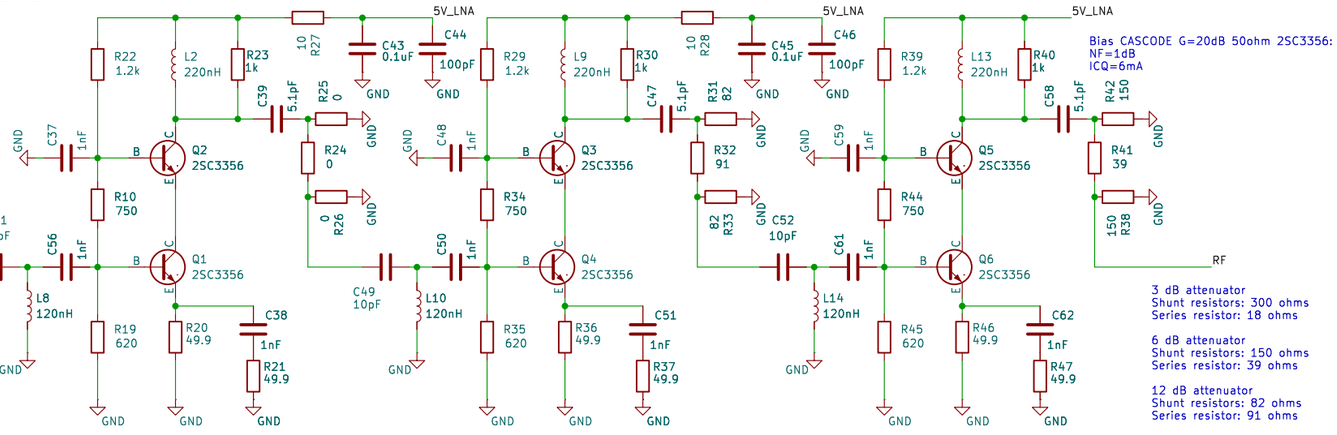
RF SHIELDING



LOW NOISE AMPLIFIER



AMPLIFIER



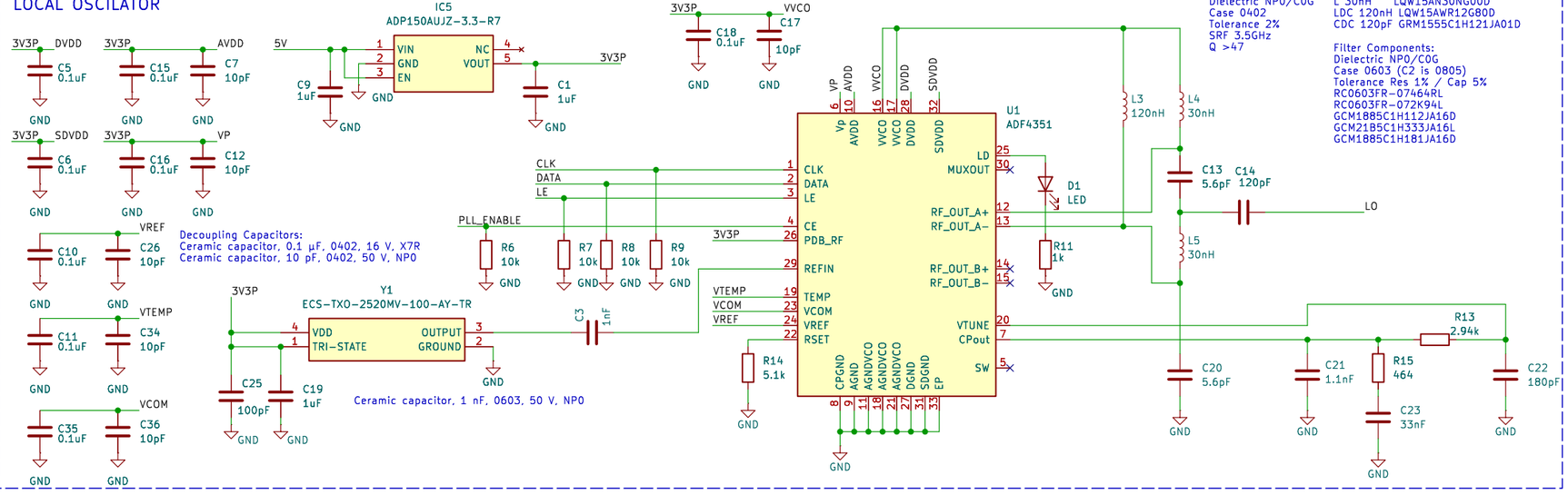
Bias CASCODE G=20dB 50ohm 2SC3356:
NF=1dB
ICQ=6mA

3 dB attenuator
Shunt resistors: 300 ohms
Series resistor: 18 ohms

6 dB attenuator
Shunt resistors: 150 ohms
Series resistor: 39 ohms

12 dB attenuator
Shunt resistors: 82 ohms
Series resistor: 91 ohms

LOCAL OSCILATOR



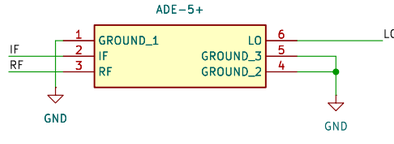
Decoupling Capacitors:
Ceramic capacitor, 0.1 µF, 0402, 16 V, X7R
Ceramic capacitor, 10 pF, 0402, 50 V, NPO

LC blaun:
Dielectric NP0/C0G
Case 0402
Tolerance 2%
SRF 3.5GHz
Q >47

C 5.6pF GJM1555C1H5R6WB01D
L 30nH LQW15AN30NG00D
LDC 120nH LQW15AWR12G80D
CDC 120pF GRM1555C1H121JA01D

Filter Components:
Dielectric NPO/C0G
Case 0603 (C2 is 0805)
Tolerance Res 1% / Cap 5%
RC0603FR-07464RL
RC0603FR-072K94L
GCM1885C1H112JA16D
GCM21B5C1H333JA16L
GCM1885C1H181JA16D

MIXER



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FCEFyN
FACULTAD DE CIENCIAS EXACTAS, FÍSICAS y NATURALES

Sheet: /
File: PCI-104-Template.kicad_sch

Title: TRANSPONDER CUBESAT VHF/UHF

Size: A3	Date: 2024-11-25	Rev: A
KiCad E.D.A. 9.0.4		Id: 1/1